

Question	1abcd	1e	2a	2bc	3	4ab	4c	5a	5bc	Total
H3	/9									/9
H5		/2	/4		/11		/4			/21
H8				/7		/7			/6	/20
H9								/5		/5
										/55

Question 1 (11 marks)

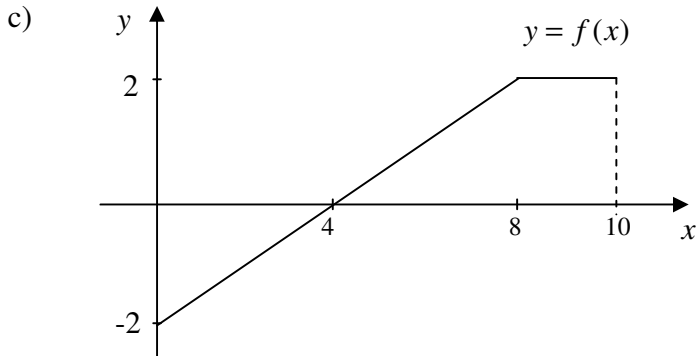
- a) If $x = 0.2$, evaluate e^{x^2} to 3 significant figures. **2**
- b) Neatly sketch $y = e^{-x}$ showing all key features. **1**
- c) Differentiate with respect to x :
- i) $y = 2e^{-x}$ **1**
- ii) $y = \frac{e^{2x}}{x+3}$ **2**
- d) Find the equation of the tangent to the curve $y = e^x + 1$ at the point $(1, e + 1)$ **3**
- e) Evaluate $\int_1^2 (x^2 + 7) dx$ **2**

Question 2 (11 marks) Start a new page.

- a) Find the indefinite integrals of
- i) $x^2(x - 3)$ **2**
- ii) $(2x + 7)^{10}$ **2**
- b) i) Sketch the curve $y = 2x^3$ in the domain $-2 \leq x \leq 2$ **1**
- ii) Find the area bounded by the curve $y = 2x^3$, the x axis, $x = 2$ and $x = -2$. **3**

Question 2 continued over page...

Question 2 continued.



Using the graph above, evaluate:

- | | | |
|------|--------------------------|----------|
| i) | $\int_4^{10} f(x) \, dx$ | 1 |
| ii) | $\int_0^4 f(x) \, dx$ | 1 |
| iii) | $\int_0^{10} f(x) \, dx$ | 1 |

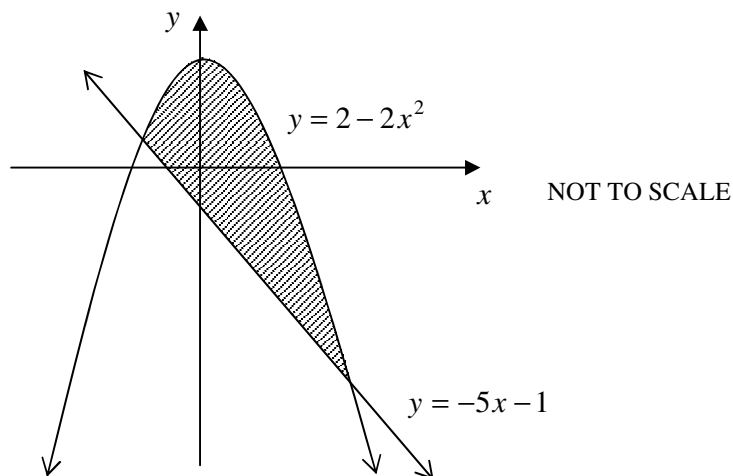
Question 3 (11 marks) Start a new page.

- | | | |
|-----|---|----------|
| a) | The first term of a geometric series is 8 and the infinite sum is 64. Find the common ratio. | 2 |
| b) | Evaluate $\sum_{n=1}^5 (-5)^{n-1}$ | 2 |
| c) | Find the sum of the first 10 terms of the series $2 + 2\sqrt{5} + 10 + \dots$ | 2 |
| d) | Mr Jordan invested \$500 at the start of this year into a bank account paying 7.5% interest per annum, compounding yearly. He plans to invest \$500 into this account at the start of every year. | |
| i) | Write an expression involving the sum of a series which will give the total value of the investment immediately after the 10 th deposit of \$500. | 2 |
| ii) | Thus calculate the balance of the bank account at this time, correct to the nearest dollar. | 1 |
| e) | By considering it as a geometric series, express $0.\dot{3}\dot{2}$ as a fraction in its simplest form. | 2 |

Question 4 (11 marks) **Start a new page.**

- a) Find the exact volume of the solid formed when the area bounded by the curve $y^2 = 8x$, the y axis and the line $y = 4$ is rotated about the y axis. **3**

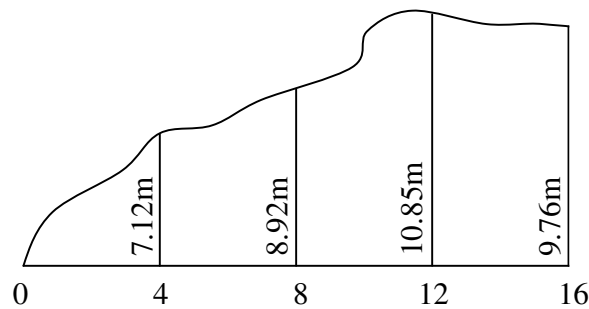
- b) The diagram below shows the graphs of the parabola $y = 2 - 2x^2$ and the straight line $y = -5x - 1$.



- i) Find the x coordinates of the points of intersection of the two curves. **1**
- ii) Find the area of the region enclosed by the two curves. **3**
- c) i) Show that $\frac{x}{(x+3)^3} = \frac{1}{(x+3)^2} - \frac{3}{(x+3)^3}$ **1**
- ii) Hence, or otherwise, evaluate $\int_0^3 \frac{x}{(x+3)^3} dx$ **3**

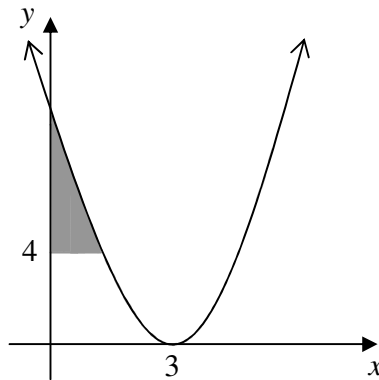
Question 5 (11 marks) **Start a new page.**

- a) Miss James is paying off her Christmas trip to Europe. She borrowed \$12 000 and is being charged 15% p.a. interest, compounding monthly. She makes equal quarterly repayments, the first being made after 3 months. Let Q be the quarterly repayment.
- i) What is the balance owing, to the nearest dollar, immediately before she makes her first repayment? **2**
- ii) If she has agreed to pay the loan off over 5 years, what is the amount of each repayment she makes? **3**
- b) Using all of the measurements below, a surveyor calculated the approximate area of this irregular block of land.



Calculate the approximate area using Simpson's rule. **3**

- c) The diagram below shows the parabola $y = (x - 3)^2$. Find the area of the shaded region. **3**



End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

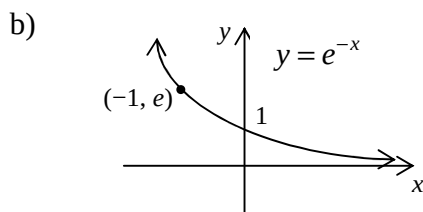
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

Solutions - Year 12 Mathematics Assessment Task 3, Term 2 2008

Question 1

a) $e^{0.04} = 1.0408107...$
 $= 1.04 \text{ (3sf)}$



c) i $y = 2e^{-x}$
 $\frac{dy}{dx} = -2e^{-x}$

ii $y = \frac{e^{2x}}{x+3}$ $u = e^{2x}$ $v = x+3$
 $u' = 2e^{2x}$ $v' = 1$

$$\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$$

$$= \frac{2e^{2x}(x+3) - e^{2x}}{(x+3)^2}$$

$$= \frac{2xe^{2x} + 5e^{2x}}{(x+3)^2}$$

d) $y = e^x + 1$ $(1, e+1)$
 $\frac{dy}{dx} = e^x$

when $x=1$, $\frac{dy}{dx} = e$, $\therefore$ gradient of tangent $= e$

$$y - y_1 = m(x - x_1)$$

$$\therefore y - (e+1) = e(x-1)$$

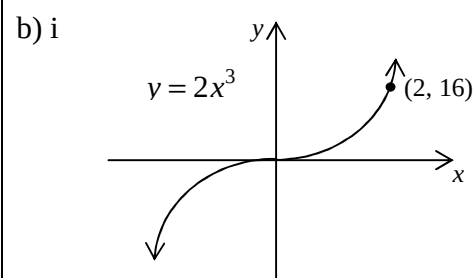
$$y = ex + 1$$

e) $\int_1^2 (x^2 + 7) dx = \left[\frac{x^3}{3} + 7x \right]_1^2$
 $= \frac{2^3}{3} + 7(2) - \left(\frac{1}{3} + 7 \right)$
 $= \frac{28}{3} = 9\frac{1}{3}$

Question 2

a) i $\int x^2(x-3) dx = \int (x^3 - 3x^2) dx$
 $= \frac{x^4}{4} - x^3 + C$

ii $\int (2x+7)^{10} dx = \frac{(2x+7)^{11}}{22} + C$



ii Function is odd, so area is symmetrical about y axis

$$A = 2 \times \int_0^2 2x^3 dx$$

$$= 2 \left[\frac{x^4}{4} \right]_0^2$$

$$= 2 \times \frac{32}{2}$$

$$= 16 \text{ units}^2$$

c) i From graph, $A = 2 \times 2 + \frac{1}{2} \times 4 \times 2$
 $= 8$

$$\therefore \int_4^{10} f(x) dx = 8$$

ii $A = \frac{1}{2} \times 4 \times 2 = 4$
 But area is below x axis
 $\therefore \int_0^4 f(x) dx = -4$

iii $\int_0^{10} f(x) dx = \int_4^{10} f(x) dx + \int_0^4 f(x) dx$
 $= 8 - 4$
 $= 4$

Question 3

a) $S_{\infty} = \frac{a}{1-r}$ $a = 8, S_{\infty} = 64$

$$64 = \frac{8}{1-r}$$

$$1-r = \frac{1}{8}$$

$$\therefore r = \frac{7}{8}$$

b) $\sum_{n=1}^5 (-5)^{n-1} \longrightarrow$ G.S. with $a = 1, r = -5, n = 5$

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$S_5 = \frac{1(1-(-5)^5)}{1+5}$$

$$= \frac{1+3125}{6}$$

$$= 521$$

c) $2 + 2\sqrt{5} + 10 + \dots \longrightarrow$ G.S with $a = 2, r = \sqrt{5}, n = 10$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$\therefore S_{10} = \frac{2\left((\sqrt{5})^{10} - 1\right)}{\sqrt{5} - 1}$$

$$= \frac{2(5^5 - 1)}{\sqrt{5} - 1}$$

$$= \frac{6428(\sqrt{5} + 1)}{4}$$

$$= 1607(\sqrt{5} + 1)$$

d) i Let A_n be the value of the investment at the start of year n .

$$A_1 = 500$$

$$A_2 = 500 \times 1.075 + 500$$

$$A_3 = 500 \times 1.075^2 + 500 \times 1.075 + 500$$

A_{10} gives the value of the investment after the 10th deposit

$$A_{10} = 500 \times 1.075^{10} + 500 \times 1.075^9 + \dots + 500$$

This is a G.S. with $a = 500, r = 1.075, n = 10$

$$\therefore A_{10} = \frac{500(1.075^{10} - 1)}{0.075}$$

ii) $A_{10} = 7073.543... \approx \7074

e) $0.\dot{3}\dot{2} = 0.32 + 0.0032 + 0.000032 + \dots$

Gives a G.S. with $a = 0.32, r = 0.01$

Since $|r| < 1$ a limiting sum exists

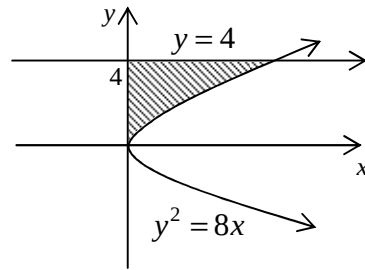
$$S_{\infty} = \frac{a}{1-r} = \frac{0.32}{1-0.01}$$

$$= \frac{0.32}{0.99}$$

$$= \frac{32}{99}$$

Question 4

a)



Volume around y axis is given by $V = \pi \int_a^b x^2 dy$

$$x = \frac{y^2}{8}$$

$$\therefore x^2 = \frac{y^4}{64}$$

$$\therefore V = \pi \int_0^4 \frac{y^4}{64} dy$$

$$= \pi \left[\frac{y^5}{320} \right]_0^4$$

$$= \frac{1024\pi}{320}$$

$$= \frac{16\pi}{5} \text{ units}^3$$

b) i Solving simultaneously,

$$2 - 2x^2 = -5x - 1$$

$$0 = 2x^2 - 5x - 3$$

$$0 = (x - 3)(2x + 1)$$

$$\therefore x = 3, -\frac{1}{2}$$

ii

$$\text{Area} = \int_{-\frac{1}{2}}^3 (2 - 2x^2) dx - \int_{-\frac{1}{2}}^3 (-1 - 5x) dx$$

$$= \int_{-\frac{1}{2}}^3 (3 - 2x^2 + 5x) dx$$

$$= \left[3x - \frac{2x^3}{3} + \frac{5x^2}{2} \right]_{-\frac{1}{2}}^3$$

$$= \left[3(3) - \frac{2(3)^3}{3} + \frac{5(3)^2}{2} \right] - \left[3\left(-\frac{1}{2}\right) - \frac{2\left(-\frac{1}{2}\right)^3}{3} + \frac{5\left(-\frac{1}{2}\right)^2}{2} \right]$$

$$= 9 - 18 + \frac{45}{2} + \frac{3}{2} - \frac{1}{12} - \frac{5}{8}$$

$$= \frac{343}{24} \text{ units}^2$$

c) i

$$\text{RHS} = \frac{1}{(x+3)^2} - \frac{3}{(x+3)^3}$$

$$= \frac{x+3-3}{(x+3)^3}$$

$$= \frac{x}{(x+3)^3} = \text{LHS}$$

ii

$$\int_0^3 \frac{x}{(x+3)^3} dx = \int_0^3 \left(\frac{1}{(x+3)^2} - \frac{3}{(x+3)^3} \right) dx$$

$$= \left[\frac{-1}{x+3} + \frac{3}{2(x+3)^2} \right]_0^3$$

$$= -\frac{1}{6} + \frac{3}{72} + \frac{1}{3} - \frac{3}{18}$$

$$= \frac{1}{24}$$

Question 5

- a) 15% p.a. = $1 \cdot 25\%$ p.m. = $0 \cdot 0125$ p.m. Principal = \$12000
 i Repayments, Q, are made at the end of every 3 months.

Let A_n be the amount owing after n months.

$$A_1 = 12000 \times 1 \cdot 0125$$

$$A_2 = A_1 \times 1 \cdot 0125$$

$$= 12000 \times 1 \cdot 0125^2$$

$$A_3 = A_2 \times 1 \cdot 0125 - Q$$

$$= 12000 \times 1 \cdot 0125^3 - Q$$

$$\therefore \text{amount owed before first payment} = 12000 \times 1 \cdot 0125^3$$

$$= 12455 \cdot 648 \dots$$

$$= \$12456 \text{ (nearest dollar)}$$

- ii Continuing the series,

$$A_4 = A_3 \times 1 \cdot 0125$$

$$= (12000 \times 1 \cdot 0125^3 - Q) \times 1 \cdot 0125$$

$$A_5 = A_4 \times 1 \cdot 0125$$

$$= (12000 \times 1 \cdot 0125^3 - Q) \times 1 \cdot 0125^2$$

$$A_6 = A_5 \times 1 \cdot 0125 - Q$$

$$= (12000 \times 1 \cdot 0125^3 - Q) \times 1 \cdot 0125^3 - Q$$

$$= 12000 \times 1 \cdot 0125^6 - 1 \cdot 0125^3 Q - Q$$

After 5 years loan is paid. 20 repayments have been made.

$$A_{60} = 12000 \times 1 \cdot 0125^{60} - 1 \cdot 0125^{57} Q - 1 \cdot 0125^{54} Q - \dots - Q$$

terms 2 onwards form a G.S. with $a = Q$, $r = 1 \cdot 0125^3$, $n = 20$

$$A_{60} = 0 \text{ since loan is repaid,}$$

$$\therefore 0 = 12000 \times 1 \cdot 0125^{60} - \frac{R(1 \cdot 0125^{60} - 1)}{1 \cdot 0125^3 - 1}$$

$$R = \frac{12000 \times 1 \cdot 0125^{60} (1 \cdot 0125^3 - 1)}{(1 \cdot 0125^{60} - 1)}$$

$$= 867.187 \dots$$

$\therefore$ quarterly repayments are \$867 (nearest dollar)

- b) 5 function values = 2 applications of Simpson's Rule

$$\int_0^{16} f(x) dx = \int_8^{16} f(x) dx + \int_0^8 f(x) dx$$

$$= \frac{8}{6} (0 + 4(7 \cdot 12) + 8 \cdot 92) + \frac{8}{6} (8 \cdot 92 + 4(10 \cdot 85) + 9 \cdot 76)$$

$$= 132 \cdot 64 \text{ m}^2$$

- c) $y = (x-3)^2$ y intercept = 9, $\therefore$ Area = $\int_4^9 x dy$

need to make x the subject: $\pm\sqrt{y} = x - 3$

$$x = 3 \pm \sqrt{y}$$

from graph, equation required is $x = 3 - \sqrt{y}$

$$\therefore A = \int_4^9 (3 - \sqrt{y}) dy$$

$$= \left[3y - \frac{2y^{\frac{3}{2}}}{3} \right]_4^9$$

$$= 27 - 18 - \left(12 - \frac{16}{3} \right)$$

$$= \frac{7}{3} \text{ units}^2$$

